Analytical Brainstorm - Interpreting Unexpected Results

Project: retail analytics eda

In this document, I reflect on why some results look the way they do. The report was created as part of a structured brainstorming session with an AI assistant, used as a thinking partner: to test ideas, challenge weak hypotheses, and separate what can be verified in the data from what cannot.

Important: the points below are HYPOTHESES, not conclusions. Where a hypothesis could be tested against the project data, it was tested and marked as verified. Where it relies on external data or assumptions, it is clearly marked as open.

Question 1: Prices rise ~37% while sales volume stays flat

Starting observation: across 2015-2024 the average unit price rises steadily (~155 to ~212, about +37%), while the total quantity sold stays flat (~161k to 174k units per year, no upward trend). A strong, sustained price increase with no reaction in volume is unusual and worth investigating: why does this happen?

Hypothesis 1 - Is it just inflation?

First idea: maybe +37% is not a real increase but simply inflation, everything got more expensive over the decade. To judge this, the price growth must be compared against an inflation benchmark for the same industry and period. The right benchmark is not random shop prices (not comparable), but an official price index for clothing and footwear.

Eurostat data - clothing & footwear inflation (HICP)

Eurostat publishes the Harmonised Index of Consumer Prices (HICP), a measure of inflation comparable across countries, broken down by category. The category "Clothing and footwear" (ECOICOP division 03) is exactly the right benchmark.

Dataset: prc_hicp_aind, measure "annual average index".

Data extracted on 04/07/2026 14:07:50
Dataset: **HICP - annual data (average index and rate of change) (1996-2025) [prc_hicp_aind__custom_22024855]**
Last updated: 06/02/2026 23:00

Time frequency: Annual
Unit of measure: Annual average index
Classification of individual consumption: Clothing and footwear

TIME	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
GEO (Labels)											
European Union (EU6-1958, EU9-1973,	99,95	100,00	100,20	100,77	100,87	101,12	102,16	103,22	106,24	110,96	112,73
European Union - 27 countries (from	100,00	100,00	100,19	100,36	100,31	100,81	100,76	101,81	104,79	109,44	111,19
European Union - 28 countries (2013-	99,95	100,00	100,20	100,77	100,87	101,12	:	:	:	:	:
Euro area (EA11-1999, EA12-2001, EA13-	99,88	100,00	100,38	100,86	100,97	101,68	101,65	102,57	104,70	108,61	110,16
Euro area – 20 countries (2023-2025)	99,88	100,00	100,38	100,86	100,96	101,67	101,63	102,55	104,71	108,64	110,19
Euro area - 19 countries (2015-2022)	99,88	100,00	100,38	100,86	100,97	101,68	101,65	102,57	104,70	108,60	110,16
Belgium	99,37	100,00	100,82	101,38	101,79	102,55	102,74	102,24	105,12	111,41	115,87

Source: Eurostat prc_hicp_aind, category CP03, EU, 2015–2024.

To convert an index to a percentage growth, we calculate it as follows: (end - start) / start × 100. For the EU: (112.73 - 100) / 100 × 100 = 12.73%. A useful shortcut: when the base equals exactly 100, the growth is simply the final value minus 100 (112.73 - 100 = 12.73%).

Interpretation: EU clothing & footwear prices rose about 12.7% over 2015-2024, while the company's average price rose about 37%, roughly THREE TIMES faster than the industry. This rules out inflation: the increase is real, above-market pricing.

Caveat: this comparison assumes the company's prices are in an EU-area currency and that its country mix is close to the EU average. A more precise version would compare per country (PL, FR, AT, IT.).

Hypothesis 2: A move to premium? (rejected)

If not inflation, is the company moving upmarket / premium? This hypothesis is weak. Mass-market clothing does not become premium simply by raising prices; historically the opposite happens (e.g. Michael Kors/Tommy Hilfiger shifted from luxury to mass-market as society grew wealthier). Premium brands also deliberately keep themselves scarce and hard to reach, which contradicts a mass brand pushing prices up across the board. So "becoming premium" does not fit.

A more plausible variant: the company may have launched a higher-positioned sub-brand or shifted its assortment upward (analogy: Reserved vs Sinsay within the same group, same owner, different positioning). This reframes the question into something testable

Hypothesis 3: Product mix vs real price increases (VERIFIED in data)

The key question became: does the average price rise because the SAME products get more expensive (real increase), or because the company sells a DIFFERENT, more expensive assortment (mix effect)? These leave two different fingerprints in the data:

- Real increase: price rises WITHIN each category over time (t-shirts, shoes, etc. all climb).
- Mix effect: price within each category stays roughly flat, but the SHARE of expensive categories grows, pulling the overall average up.

Since the analysis view did not contain the product category, it was first added by joining products_mmmgmeum and re-defining the view:

```
ALTER VIEW seasonality_VS_margin AS
SELECT sales_orders.product_id, category, order_date, quantity, unit_price,
       base_price, discount_pct, unit_price*quantity AS revenue, status
FROM sales_orders
INNER JOIN products_mmmgmeum ON sales_orders.product_id = products_mmmgmeum.product_id;
```

Then the average price was calculated per category and per year:

```
SELECT category, YEAR(order_date) AS year, ROUND(AVG(unit_price),2) AS AVG_price
FROM seasonality_VS_margin
GROUP BY category, YEAR(order_date)
ORDER BY category, YEAR(order_date);
```

CATEGORY	YEAR	AVG_PRICE	CATEGORY	YEAR	AVG_PRICE	CATEGORY	YEAR	AVG_PRICE	CATEGORY	YEAR	AVG_PRICE	CATEGORY	YEAR	AVG_PRICE
Accessories	2015	120,48	Kids	2015	90,85	Men	2015	161,82	Shoes	2015	224,00	Women	2015	183,67
Accessories	2016	128,17	Kids	2016	94,00	Men	2016	171,04	Shoes	2016	232,34	Women	2016	191,28
Accessories	2017	131,43	Kids	2017	98,22	Men	2017	175,70	Shoes	2017	241,25	Women	2017	198,65
Accessories	2018	137,03	Kids	2018	101,33	Men	2018	180,98	Shoes	2018	251,26	Women	2018	205,31
Accessories	2019	141,61	Kids	2019	105,82	Men	2019	188,25	Shoes	2019	260,99	Women	2019	212,87
Accessories	2020	145,62	Kids	2020	109,89	Men	2020	195,02	Shoes	2020	269,47	Women	2020	220,64
Accessories	2021	150,51	Kids	2021	113,03	Men	2021	202,12	Shoes	2021	280,18	Women	2021	226,21
Accessories	2022	156,11	Kids	2022	116,27	Men	2022	206,63	Shoes	2022	287,01	Women	2022	235,60
Accessories	2023	160,21	Kids	2023	119,96	Men	2023	213,49	Shoes	2023	297,24	Women	2023	244,11
Accessories	2024	164,71	Kids	2024	123,25	Men	2024	220,57	Shoes	2024	305,07	Women	2024	249,74
	%	**36,71**		%	**35,66**		%	**36,31**		%	**36,19**		%	**35,97**

The table shows the percentage growth of the average price by category over the years 2015–2024.

Result: The price rises steadily within the categories checked (Accessories +37%, Kids +35% over 2015–2024), at roughly the same pace as the overall average. The full-population test below (Check D) confirms this holds across all products."

In combination with Eurostat data (industry +12,7%), this data suggests that the company's revenue growth is driven by an aggressive, above-market pricing strategy.

Hypothesis 4: deeper check (does the category average hide a within-category mix?)

A category average can still hide a mix effect one level down: the average of "Shoes" could rise not because existing shoes get pricier, but because expensive new shoe models join the category over time. Three additional checks were run to rule this out.

Check A: do the same individual products get more expensive over time?

Average price was tracked per product per year for a sample of 9 products (3 each from Kids, Men, Women), all present since 2015:

```
SELECT category, product_id, YEAR(order_date) AS year, ROUND(AVG(unit_price),2) AS AVG_price
FROM seasonality_VS_margin
WHERE product_id IN ( ... )
GROUP BY category, product_id, YEAR(order_date)
ORDER BY category, product_id, YEAR(order_date);
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N
CATEGORY	PRODUCT_ID	YEAR	AVG_UNIT PRICE		CATEGORY	PRODUCT_ID	YEAR	AVG_UNIT PRICE		CATEGORY	PRODUCT_ID	YEAR	AVG_UNIT PRICE
Kids	114	2015	135,93		Men	68	2015	196,81		Women	253	2015	210,88
Kids	114	2016	137,13		Men	68	2016	207,44		Women	253	2016	207,53
Kids	114	2017	139,92		Men	68	2017	233,28		Women	253	2017	214,34
Kids	114	2018	140,07		Men	68	2018	219,32		Women	253	2018	232,21
Kids	114	2019	142,96		Men	68	2019	249,25		Women	253	2019	236,38
Kids	114	2020	151,67		Men	68	2020	230,54		Women	253	2020	249,93
Kids	114	2021	158,76		Men	68	2021	255,24		Women	253	2021	239,56
Kids	114	2022	166,97		Men	68	2022	251,73		Women	253	2022	268,57
Kids	114	2023	169,36		Men	68	2023	257,53		Women	253	2023	260,89
Kids	114	2024	170,22		Men	68	2024	275,27		Women	253	2024	260,21
		%	25,23				%	39,87				%	23,39
Kids	710	2015	73,11		Men	802	2015	106,55		Women	1637	2015	182,5
Kids	710	2016	79,37		Men	802	2016	114,43		Women	1637	2016	189,58
Kids	710	2017	78,54		Men	802	2017	117,76		Women	1637	2017	207,1
Kids	710	2018	85,78		Men	802	2018	120,36		Women	1637	2018	221,4
Kids	710	2019	88,06		Men	802	2019	123,85		Women	1637	2019	216,45
Kids	710	2020	88,54		Men	802	2020	131,36		Women	1637	2020	220,61
Kids	710	2021	99,79		Men	802	2021	132,85		Women	1637	2021	247,19
Kids	710	2022	91,16		Men	802	2022	136,56		Women	1637	2022	242,3
Kids	710	2023	97,84		Men	802	2023	142,05		Women	1637	2023	238,81
Kids	710	2024	103,17		Men	802	2024	151,9		Women	1637	2024	230,05
		%	41,12				%	42,56				%	26,05
Kids	2459	2015	148,38		Men	2059	2015	127,69		Women	2094	2015	211,62
Kids	2459	2016	138,65		Men	2059	2016	148,39		Women	2094	2016	223,37
Kids	2459	2017	151,81		Men	2059	2017	136,47		Women	2094	2017	230,83
Kids	2459	2018	161,62		Men	2059	2018	146,05		Women	2094	2018	232,82
Kids	2459	2019	158,65		Men	2059	2019	157,42		Women	2094	2019	244,72
Kids	2459	2020	174,17		Men	2059	2020	162,35		Women	2094	2020	248,1
Kids	2459	2021	179,67		Men	2059	2021	158,04		Women	2094	2021	260,57
Kids	2459	2022	170,92		Men	2059	2022	172,12		Women	2094	2022	271,47
Kids	2459	2023	172,99		Men	2059	2023	178,76		Women	2094	2023	282,13
Kids	2459	2024	185,44		Men	2059	2024	183,77		Women	2094	2024	286,36
		%	24,98				%	43,92				%	35,32

The table shows the percentage growth of the average price between 2015 and 2024 for randomly selected products and categories.

Every one of the 9 products rose in price from 2015 to 2024, by +23% to +44% (average ~34%), with mild year-to-year fluctuations. The SAME products genuinely get more expensive over time. (Note: this is an illustrative sample of self-selected products, not a full-population proof.)

Check B: New products appear cyclically - are they all from 2015?

Product count by launch year was checked in products_mmmgmeum:

```
SELECT YEAR(launch_date) AS year, COUNT(product_id) AS product_count
FROM products_mmmgmeum
GROUP BY YEAR(launch_date);
```

	year	product_count
1	2024	233
2	2021	260
3	2022	239
4	2016	253
5	2019	227
6	2020	234
7	2023	240
8	2017	226
9	2018	236
10	2015	248

A table showing the number of new products introduced in the years 2015–2024

Roughly 226-260 new products are introduced every year across the whole period. The assortment renews itself continuously, which is realistic and a good sign about data quality (if all products had a 2015 launch date, the dataset would look artificial).

Check C: do newer products enter at higher prices?

Average catalogue price (base_price) was calculated per launch year:

```
SELECT YEAR(launch_date) AS year, ROUND(AVG(base_price),2) AS AVG_PRICE
FROM products_mmmgmeum
GROUP BY YEAR(launch_date);
```

	year	AVG_PRICE
1	2024	156.120000
2	2021	155.220000
3	2022	151.120000
4	2016	156.500000
5	2019	149.900000
6	2020	158.090000
7	2023	147.910000
8	2017	152.410000
9	2018	155.540000
10	2015	157.130000

A table showing the average price for products introduced in a given year.

Result: the average catalogue price of newly launched products is flat across launch years (~150-158; the 2024 cohort at 156 is essentially the same as the 2015 cohort at 157). Newer products do NOT enter at higher prices, which rules out the "expensive new products" explanation.

Conclusion of the deeper check: the three tests together confirm that the rising average price reflects real price increases on existing products, not a change of assortment. Interestingly, catalogue prices at launch stay flat over the years, while the actual selling price of a given product rises during its life so the increases happen during a product's lifetime, not at launch.

Check D: full-population confirmation (all products present in both 2015 and 2024)

Finally, to move beyond the illustrative sample, the price change was calculated for EVERY product present in both years, by joining a 2015 price snapshot and a 2024 price snapshot on product_id (an INNER JOIN keeps only products present in both years):

```
SELECT p2015.product_id, p2015.AVG_unit_price_2015, p2024.AVG_unit_price_2024,
  (p2024.AVG_unit_price_2024 - p2015.AVG_unit_price_2015)
   / p2015.AVG_unit_price_2015 * 100 AS price_growth_pct
FROM ( SELECT product_id, ROUND(AVG(unit_price),2) AS AVG_unit_price_2015
       FROM seasonality_VS_margin WHERE YEAR(order_date) = 2015
       GROUP BY product_id ) AS p2015
INNER JOIN ( SELECT product_id, ROUND(AVG(unit_price),2) AS AVG_unit_price_2024
             FROM seasonality_VS_margin WHERE YEAR(order_date) = 2024
             GROUP BY product_id ) AS p2024
ON p2015.product_id = p2024.product_id;
```

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	product_id	AVG_unit_price_2015	AVG_unit_price_2024	price_growth_pct
1	13	119.960000	155.020000	29.226400
2	15	161.440000	211.950000	31.287100
3	30	91.130000	127.970000	40.425700
4	32	175.190000	224.840000	28.340600
5	47	111.280000	157.940000	41.930200
6	49	277.420000	395.110000	42.423000
7	64	121.880000	159.140000	30.571000
8	66	232.130000	312.590000	34.661600
9	81	118.700000	170.220000	43.403500
10	83	235.940000	326.090000	38.208800
11	98	264.230000	363.300000	37.493800
12	100	169.060000	212.140000	25.482000
13	113	124.700000	177.780000	42.566100
14	115	98.570000	137.000000	38.987500
15	130	245.450000	319.620000	30.217900
16	132	73.310000	90.230000	23.080000
17	147	196.160000	280.890000	43.194300
18	164	60.360000	82.350000	36.431400
19	166	147.960000	198.830000	34.380900
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

A comparison of the average price growth for all products between 2015 and 2024.

Result: of the 2,396 products present in both years, the average price growth is 36.0% (median also 36.0%), ranging from +8.8% to +73%. Not a single product became cheaper, and only 3 rose by less than 10%. This confirms across almost the entire product population what the 9 product sample suggested: existing products systematically and substantially increased in price over the decade.

Data-quality caveat, is the volume real?

The volume staying almost perfectly flat for a decade, despite continuous price increases, is itself suspicious price and demand are usually linked. Before drawing conclusions about demand, it would be worth verifying that the quantity data is collected correctly.

Why does volume stay flat? (open hypotheses)

If the product sells well and is profitable, it would be natural to grow volume the fact that it does not suggests some constraint. Possible explanations:

- Production limits - no spare capacity, or investment in a new line has not (yet) paid off.
- Deliberate scarcity - keeping supply lower to protect a higher margin per unit (not necessarily a mistake possibly a conscious choice).
- Market saturation - no new customers without expanding into new markets.
- A deliberate "fewer, pricier, higher-margin" business model.

Caveat: none of these hypotheses can be resolved with the current data. To do so, I would need information on, among other things, costs, production capacity, and the market. It is important NOT to assume a "bad strategy": stable volume with rising prices and profits can be a rational, deliberate choice.

Question 1: summary

- The +37% price rise is real, not inflation: EU clothing inflation was only 12,7% over the same period (Eurostat).
- It is not a mix effect confirmed four ways:
    - a) the same individual products rise +23% to 44% over time;
    - b) products enter continuously every year (~240/year), so the data is realistic;
    - c) newly launched products do not enter at higher prices (base_price flat ~150 to 158);
    - d) across the full population, all 2,396 products present in both 2015 and 2024 rose in price, by 36% on average (none became cheaper).
- The increases happen during a product's lifetime, not at launch (catalogue prices at launch stay flat, actual selling prices of existing products rise).
- Overall: an above-market pricing strategy (~3× faster than the industry), applied to existing products.
- Flat volume despite rising prices is unusual, quantity data should be validated, and several supply/demand hypotheses remain open (not resolvable with current data).

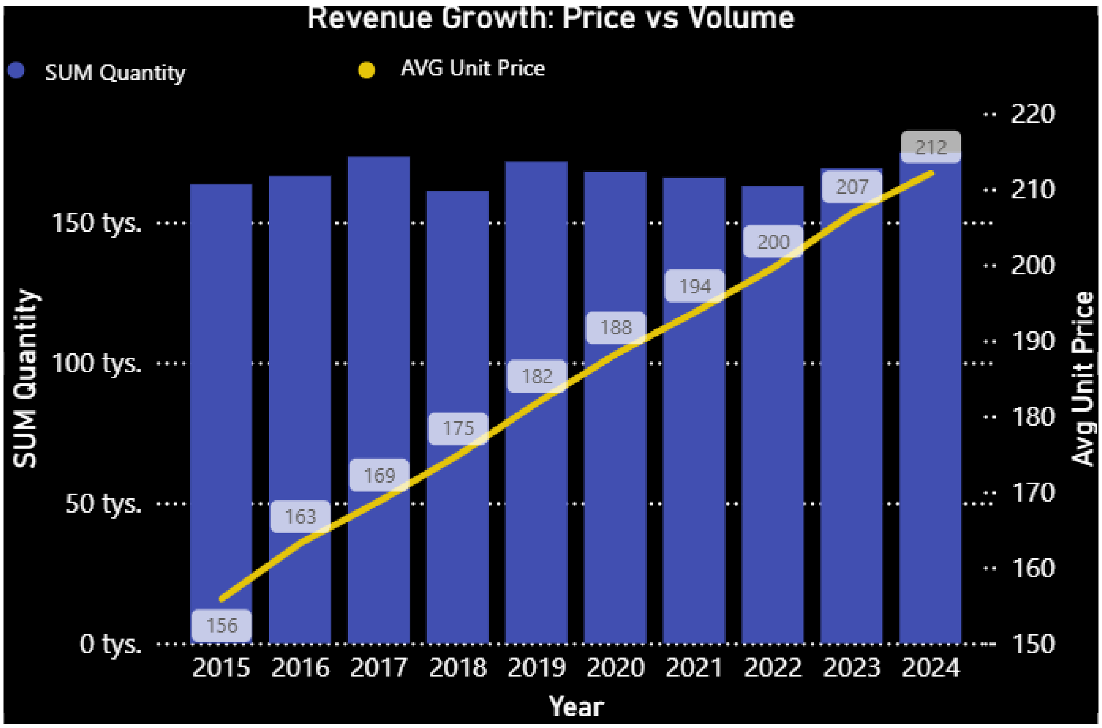
Question 2: Revenue stalls in 2018 and dips in 2022

Observation: on total yearly revenue, the otherwise steady growth flattens or reverses twice. Revenue barely moves from 2017 to 2018 (~28.9M to 29.0M, +0.1M) and falls from 2021 to 2022 (~32.4M to 32.1M, the only decline in the decade).

A chart showing revenue from 2015 to 2024

Step 1: which component weakened: price, orders, or units?

Since revenue = price × quantity, and price is known to rise every year, a revenue stall must mean quantity fell. The price-vs-volume chart confirms it: the average price keeps rising, but the quantity bars dip in exactly 2018 and 2022.



A chart comparing the number of products sold with their average price in a given year.

To separate "fewer orders" from "fewer units per order", the number of orders and total units were compared per year (a prior check confirmed each order contains exactly one product, so COUNT of rows equals the number of orders):

```
SELECT YEAR(order_date) AS year, COUNT(order_id) AS num_of_orders,
       SUM(quantity) AS quantity_a_year
FROM sales_orders
GROUP BY YEAR(order_date);
```

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	year	NUM_of_orders	Quantity_a_year
1	2023	25313	175393
2	2019	25469	179175
3	2018	25258	169992
4	2016	25632	173463
5	2024	25711	180718
6	2022	25689	167421
7	2017	25509	180504
8	2015	25375	169240
9	2021	25455	175889
10	2020	25393	173748

A comparison of the number of placed orders with the total number of ordered units from 2015 to 2024.

Result: the number of orders is essentially flat every year (~25,300–25,700), including 2018 and 2022. What drops is the number of units: 2018 falls to ~169,992 (from ~180,504 in 2017) and 2022 is the lowest of the decade at ~167,421. So customers placed the usual number of orders but bought fewer units per order (roughly 7.1 to 6.7 units in 2018). The revenue stall is a volume effect, not a price or order-count effect

Step 2: interpreting the two dips

2022 has a clear external explanation: the war in Ukraine starting in February 2022, the energy shock, and the surge in inflation hit European trade hard (and this data is European: PL, FR, AT, IT.). Fewer units per order is consistent with customers cutting spending during a crisis; the fact that 2022 has the lowest volume is coherent.

2018 has no obvious external cause it was, according to most sources, a good year economically. The drop is small: from 7.1 to 6.7 units per order. It remains unexplained. Possible causes: an internal company factor not visible in the data, or an artifact of the training dataset. Rather than forcing an explanation, 'I'm leaving this unresolved rather than guessing

A positive observation worth noting: during the pandemic, the company did NOT record a decline 2020 and 2021 grew, just more slowly. The company proved resilient (slower growth, no decline) that period, for example a shift to online sales.

Question 2: summary

- Revenue stalls in 2018 and dips in 2022; in both cases the cause is a fall in units sold, not price or order count (verified).
- 2022 fits the external shock (war in Ukraine, energy crisis, inflation) on the European market.
- 2018 is a real but small, unexplained dip. Left honestly as unresolved / possible noise in a practice dataset.
- The company showed resilience through the pandemic - slower growth, but no decline.

Next steps

Question 1 and Question 2 are covered. Remaining hypotheses about the flat volume (production capacity, deliberate scarcity, market saturation) stay open, as they cannot be resolved with the current data. Any further work would benefit from external context (costs, capacity, market share) that this dataset does not contain.